

BUTLERS GORGE, AustraliaWMO: **949590**Lat: **42.2753S**Lon: **146.2758E**Elev: **2188**StdP: **13.57**Time Zone: **10.00 (AUS)**Period: **96-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.3	26.8	20.1	16.3	35.6	23.2	18.9	31.6	28.0	36.6	24.9	33.7	3.7	240	0.641

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	21.5	78.8	58.8	74.6	57.8	71.1	56.7	61.7	72.9	59.9	70.3	58.2	67.8	8.6	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
58.1	78.2	64.4	56.0	72.3	62.7	53.9	67.1	61.4	28.6	73.3	27.3	70.5	26.1	68.2	70.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.9	21.0	17.5	21.8	86.2	1.4	4.9	20.8	89.7	20.0	92.6	19.2	95.3	18.2	98.8		
(5)			20.9	65.3	1.1	2.3	20.1	67.0	19.5	68.3	18.9	69.6	18.1	71.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	45.6	54.6	54.1	51.2	45.9	42.1	39.8	38.1	38.8	41.4	44.1	47.6	50.5		
	DBStd	7.98	6.70	6.60	6.74	5.40	4.64	3.93	4.01	4.02	4.83	5.88	6.92	6.91		
	HDD50	2203	27	31	68	142	247	306	368	348	261	201	125	79		
	HDD65	7079	329	309	430	572	709	756	833	813	708	648	521	451		
	CDD50	612	170	145	104	20	3	0	0	0	4	18	55	93		
	CDD65	14	8	3	2	0	0	0	0	0	0	0	0	1		
	CDH74	432	212	74	42	0	0	0	0	0	0	5	31	68		
	CDH80	90	57	11	8	0	0	0	0	0	0	0	3	11		
	WSAvg	6.7	6.4	7.1	6.3	6.6	6.9	5.3	6.2	6.2	6.8	7.5	7.2	7.5		
	PrecAvg	64.9	3.9	3.0	3.7	5.7	6.2	5.6	6.9	6.8	6.6	6.3	5.4	5.0		
(15) Precipitation	PrecMax	89.8	8.8	7.3	8.5	13.6	16.8	15.9	15.3	13.0	17.8	22.3	15.1	10.5		
	PrecMin	42.8	1.3	0.7	1.1	2.0	1.5	2.2	3.0	1.7	1.5	1.7	1.5	0.2		
	PrecStd	11.4	1.9	1.7	1.7	2.6	3.3	2.8	3.0	3.1	3.0	3.5	2.5	2.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	86.3	81.1	79.4	67.0	59.3	53.1	51.2	55.5	62.6	72.6	78.2	81.0		
		MCWB	62.1	60.9	60.3	55.2	50.0	48.4	45.7	47.5	51.0	55.1	57.4	58.4		
	2%	DB	79.8	75.6	72.4	62.7	54.9	50.9	48.6	51.7	57.0	65.4	72.0	74.6		
		MCWB	59.6	59.1	58.3	52.8	49.0	46.6	44.4	45.4	48.0	50.9	55.7	56.5		
	5%	DB	74.4	72.0	68.2	59.2	52.5	49.0	47.1	48.8	53.6	60.9	66.9	69.2		
		MCWB	57.8	57.7	56.9	51.7	48.1	45.4	43.4	44.0	46.3	50.2	53.4	55.0		
	10%	DB	69.8	67.8	63.9	55.9	50.1	47.4	45.4	46.3	50.5	56.6	61.6	64.8		
		MCWB	56.3	56.4	55.4	50.2	46.4	44.3	42.2	42.8	45.1	48.5	51.7	54.2		
	0.4%	WB	65.9	63.4	62.4	57.5	51.9	49.7	47.5	48.4	51.7	56.5	59.5	62.0		
		MCDB	79.3	73.6	73.7	64.7	56.0	51.7	49.7	53.2	60.6	64.3	71.7	71.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	61.7	61.2	60.0	54.7	49.9	47.4	45.0	46.4	48.9	53.4	57.6	58.8		
		MCDB	75.1	70.6	69.1	60.5	53.5	49.9	47.8	50.9	54.8	62.8	68.5	69.1		
	5%	WB	59.2	59.5	58.1	52.5	48.3	45.9	43.7	44.6	47.2	50.9	55.3	56.6		
		MCDB	71.5	68.4	65.9	57.7	51.7	48.5	46.5	48.1	52.2	59.3	64.5	67.0		
	10%	WB	57.1	57.3	56.0	50.6	46.8	44.5	42.5	43.1	45.6	48.5	52.5	54.6		
		MCDB	67.6	65.8	63.4	55.5	50.0	47.1	45.3	46.1	50.1	56.4	61.1	64.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	21.5	20.7	18.2	15.6	12.1	11.9	11.2	12.2	14.2	17.9	18.7	19.4		
		MCDBR	33.1	28.8	27.9	23.6	17.8	15.7	14.6	18.1	21.4	27.9	30.3	30.7		
		MCWBR	15.3	14.4	14.8	14.6	12.5	12.2	11.0	12.5	13.2	16.0	15.6	15.3		
	5% WB	MCDBR	28.6	24.5	24.3	21.1	15.6	13.5	13.1	16.6	18.3	25.3	27.1	26.2		
		MCWBR	14.7	14.1	14.0	13.9	11.7	10.9	10.3	11.9	12.2	15.5	15.2	14.4		
(40) Clear-Sky Solar Irradiance	taub		0.312	0.305	0.301	0.292	0.279	0.269	0.271	0.290	0.307	0.309	0.307	0.306		
	taud		2.587	2.624	2.624	2.626	2.642	2.653	2.653	2.594	2.534	2.545	2.579	2.597		
	Ebn at Noon		322	317	304	285	268	260	268	281	296	311	322	326		
	Edh at Noon		32	30	27	23	19	17	18	23	29	32	32	32		
(44) All-Sky Solar Radiation	RadAvg		2057	1780	1310	887	537	425	476	712	1052	1516	1864	2011		
	RadStd		157	111	78	58	42	38	30	57	69	98	150	140		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (34 neighbors)	+0.58	N/A	N/A	+1.21	+0.59	N/A	N/A	-176	+191	+22

Nomenclature: See separate page